Smart Dam Alert System with Automated Gate Control

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1. Introduction

Floods in the Puttalam region of Sri Lanka occur because of the unavailability of automation in dam monitoring. Manually operated gate responses take about 30+ minutes, and may lead to destruction of property, losses in agriculture, and even human lives. SDAS addresses the issue by integrating IoT sensors, machine learning-based prediction, threshold mechanism in the controller, a cloud backend using Supabase, and two mobile applications – one for public and another for operators.

1.1 Summary of literature

Weerasinghe et al. [1] explored the characteristics necessary for designing an EWS to ensure minimal chances of dam failures in Sri Lanka. They have created the design criteria of the local dams' safety system but failed to develop the prototype of the automated system. Sasikala et al. [2] created the IoT-based system for the water level monitoring and management in reservoirs that proved to be effective with regards to real-time sensor acquisitions but lacked automatic gate control and warning system features for the end users. Nirmal and Shekapure [3] suggested an AI-based intelligent management system of dams for ensuring safety and flood protection using the machine learning techniques but did not consider the multi-layered public alert process.

1.2 Research Gap

In the examined literature, the systems can be grouped into those that

(a) have water level monitoring while gate operation and warnings are left to manual actions,

(b) use automation of gate operation but only have general SMS notifications without having a mobile application that can reach both users at two levels, and (c) use artificial intelligence/machine learning for decision-making while lacking a three-stage warning protocol. None of the present systems includes

(i) IoT-based water level monitoring in real-time,

(ii) machine learning-based water level forecast and sensor anomalies detection,

(iii) automated gate operation with manual override,

(iv) three-stage warning protocol including pre-warning, clear area warning, and danger phases, and

(v) two separate mobile applications: public (login-free) and operator (login-based).

1.3 Problem definition

The manual manipulation of the gates leads to a delay of more than 30 minutes during emergencies; the lack of monitoring of the water level prevents any early warning system; there is only a small group of people who can be alerted using the radio and telephones; there is no separate mobile-based interface for the general public and the operators; there is no gradual warning system to alert the communities surrounding the area before the full opening of the gate; and there is no automated decision-making process with the help of machine learning.

1.4 Research Questions

(1) How can one guarantee optimal water level sensing with IoT sensors in a resilient fashion?

(2) What kind of logic should be safe to apply for total automation of the gate operation yet allow manual control as well?

(3) What is the precision rate of SMS and local alarms in notifying emergency services in 2 seconds?

(4) Is it possible to create an app for mobile devices that will allow operators of the flood to receive real-time monitoring data and make decisions accordingly?

(5) How precise can a machine learning model be in predicting rising water levels and detecting sensor malfunctions to implement a multi-stage public warning system?

2. Goal of the project

Design a prototype of a Smart Dam Alarm System that incorporates IoT devices, embedded software, Supabase cloud services, machine learning algorithms, SMS alarms, and two different mobile applications. This system offers real-time water level monitoring, automatic gate regulation, public alarms, and operator manual override of system operation. There are two mobile applications – one for the community (without logging in) and another for the operator (with login). The LSTM model is used to predict the water level in 1 hour, while the Autoencoder algorithm is applied to detect anomalies in sensors' readings.

3. Aims and objectives

- Design an end-to-end IoT-based smart dam monitoring system comprising of hardware (ESP32, water level and temperature sensors, gate motor, buzzer, RGB LED), embedded software, cloud database (Supabase), and mobile applications.
- Design two React Native (Expo) applications: Public App for real-time water level updates and notifications, and Operator App that provides authentication, gate controlling, and alert handling capabilities.
- Apply machine learning (TensorFlow) techniques using LSTM model for predicting water level for the short term and Autoencoder for detecting anomalies.
- Send staged notifications using SMS through the SIM800L GSM module when water levels increase and gates open.
- Test the system with high precision (±2 cm), low latency (<2 s), reliable gate operation (>100 cycles), and more than 95% SMS delivery rate.
- Ensure reliability through use of temperature-compensated double ultrasonic sensing technology, hysteresis window (3%), and edge machine learning.

4. Expected Outcomes

Deliverables:

- Fully working IoT hardware prototype (with sensors and motors)
- C/C++ firmware (with automated decision making < 2s)
- Supabase (PostgreSQL) database (for storing sensors' readings, alarms and contacts)
- Released public mobile application (without login) (visualisation of sensor values and alarm notifications)
- Operator mobile application (with login) (manually controlling the gate, viewing the history of alarms)
- Machine learning model (for water level prediction and detection of anomalies in sensor readings)
- Staged SMS alarm notifications system (pre-alarm notification when 20-30% gate is opened, clear area alarm and danger alarm)
- Test report
- Hybrid LSTM+Autoencoder ML pipeline for one hour ahead prediction and anomalies detection, run on the edge-inference server
- Detailed technical documentation

Status	Trigger	Gate	Colour	Alert Sent
NORMAL	<70% capacity	Closed	Green	No alert; continuous monitoring
PRE-WARNING	70-85%; gate opens 20-30%	Opening 20-30%	Yellow	Pre-warning SMS to all registered contacts
CLEAR-AREA WARNING	Rising further; gate 30-70%	Progressively opening	Orange	"Clear the area" SMS to all registered contacts
DANGER	>85%; gate beyond warning threshold	Auto full open	Red	Local alarm + danger SMS to all operators & public list

Operator override is accessible to the operators through the operator mobile application always. Criteria for evaluation include sensor accuracy, sensor response time, gate cycle reliability, rate of delivery of the staged SMS alerts, machine learning prediction accuracy and false positive rate, and mobile application data-refresh time.

Accuracy of ±2.0cm is expected from the sensors, sensor to gate response time of <2.0s, gate cycles 100+, SMS delivery rate of 95%+, and <1.0s mobile application data-refresh time. For the LSTM model, MAPE of <5% is targeted for 1-hour ahead prediction, Autoencoder will detect anomalies of the sensors within 5 seconds, and SIM800L SMS broadcast will take less than 10 seconds for reaching the local emergency contact list.

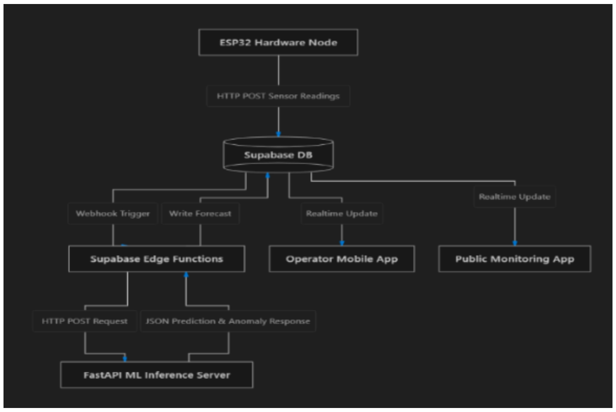
5. Proposed methodology

There are five parts of the system namely: Hardware (HC-SR04 ultrasonic sensor, temperature sensor, 12V DC motor with relay, SIM800L GSM, buzzer, RGB LED); Microcontroller (ESP32 with C/C++ software that will read the sensors, apply thresholds, control motors and run offline); Cloud (Supabase backend, PostgreSQL database); Machine Learning (TensorFlow model using past water level and rainfall to predict water levels in the next hour and detect anomalies) and Mobile Applications (Public app without any need for login, Operator app with a secure login). The ML results feed into the decision making regarding the gate operation as a recommendation rather than a trigger itself. An LSTM component will forecast water level 1 hour ahead based on the historical data of the sensor while the Autoencoder will reconstruct the data coming from the sensors and calculate MSE between the reconstruction and the original data, any reading above a certain threshold will be considered an anomaly. Hysteresis is used for all status changes at 3%.

6. Resource requirements

ESP32, HC-SR04, DHT11/22, 12V DC motor, relay module, SIM800L GSM, buzzer, RGB LED, 12V lithium battery (if necessary) are among the hardware to be used. Software: Arduino IDE (C/C++), Supabase (PostgreSQL, Auth, Edge Functions), React Native with Expo (two mobile applications), TensorFlow (ML forecasting/anomaly detection model), Git, and GitHub. Cloud Services: Supabase Cloud (free tier); SMS services: local SIM800L gateway (paid). Dependencies to be considered: Information on water resources of the

Puttalam District, flood history data for 2017-2023, dam design information, and sensor calibration standards



SDAS data flow: ESP32 hardware node posts sensor readings to the Supabase database, which triggers Edge Functions for ML inference (FastAPI server) and pushes realtime updates to the operator and public mobile applications

7. Ethics and AI Compliance

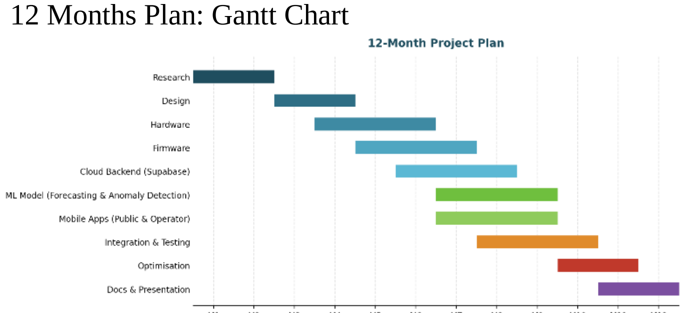
No personal information from human subjects is used in the process of training the models with machine learning. The models are trained on the historical hydrological and sensor data only. Contact information for operators and the general public is kept safely in Supabase after being encrypted for SMS emergency alerts. Operators' authentication is performed through Supabase Auth and accessing the database is done by selected administrators only. The machine learning model predictions and anomalies detection provide the input into a rules-based decision system, while gate automation happens according to the pre-defined safety thresholds..

8. Risk Assessment

Technical: sensor problem (double sensor redundancy); network problem (ESP32 offline operation and backup through GSM); motor problem (manually operable); ML risk of false predictions or false positives for anomalies (offset with having the ML outputs only advisory at a rules-based thresholding layer). Operational: delay in schedule (12-month buffer plan); resource unavailable (early acquisition); mobile app store approval (early submission and Expo). Project: scope creep (scope defined, in alignment with supervisor); integration difficulties (modular approach, continuous testing).

09. Timeplan

12 Months Plan: Gantt Chart



10. Conclusion

SDAS is an end-to-end flood management system that includes hardware, firmware, cloud-based backend (using Supabase), ML-based forecast and anomalies prediction capabilities, and two mobile applications (for public and operators). All of this is interconnected and designed for use within the district of Puttalam. Restrictions: prototype level deployment only. Further research will include integration with weather predictions, connection to the disaster management systems of Sri Lanka, and prediction capabilities using ML.

11. References

The IEEE referencing format (provided below) is used for referencing.

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